EXECUTIVE RESEARCH & DATA PIPELINE REPORT

North Bengal Groundwater Hydrochemistry & Smart Heavy-Metal Screening

EXECUTIVE SUMMARY

This report provides a clear, step-by-step overview of the groundwater quality analysis and smart screening model developed for the North Bengal aquifer (N = 40 regional wells).

The primary objective is to enable **rapid, low-cost field screening** for dangerous heavy metals (**Cadmium** and **Nickel**) using 4 routine water parameters (**pH, TDS, Nitrate, and Well Depth**), significantly reducing the need for immediate, expensive laboratory analytical testing.

1. STEP-BY-STEP DATA PREPARATION PIPELINE

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[ STEP 1: RAW DATA COLLECTION ]
  • Excel File: Groundwater_quality_data_Northbengal.xlsx (N = 40 tube wells)
  • Parameters: Major cations (Ca, Mg, Na, K), anions (Cl, HCO3, SO4, NO3), and trace metals (As, Pb, Ni,
         ↓
[ STEP 2: QA/QC & MISSING DATA IMPUTATION ]
  • Ionic Balance Check: Confirmed cation-anion balance error (CBE) < 5% across wells
  • Sample S98_01798 had 1 missing Well Depth value
  • Treatment: Imputed via median depth (22.0 m) in Scikit-Learn Pipeline
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[ STEP 3: HYDROCHEMISTRY & RISK INDEXING ]
  • Evaluated water usability: Water Quality Index (WQI)
  • Evaluated heavy metal pollution: Heavy Metal Pollution Index (HPI) & Heavy Metal Evaluation Index (HEI
  • Classified hydrochemical facies: Gibbs Mechanisms & Ion Exchange Processes
         ↓
[ STEP 4: FINAL ML FEATURE DATASET CREATION ]
  • File Created: data/processed/phase4_features_track1.csv
  • Standardized feature matrix ready for machine learning model training (N = 40 rows)
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2. INPUT FEATURES VS. OUTPUT TARGETS

Category	Parameter Name	Column Code in Dataset	Unit	Role / Purpose
Input Feature 1	Water pH	`pH_proxy`	pH unit	Controls metal solubility & mobility
Input Feature 2	Total Dissolved Solids	`TDS_calc`	mg/L	Indicates overall mineral concentration
Input Feature 3	Nitrate-Nitrogen	`NO3-N_num`	mg/L	Proxy for agricultural/anthropogenic contamination
Input Feature 4	Well Depth	`WELL_DEPTH`	meters	Captures aquifer depth layer
Output Target 1	Nickel Concentration	`Ni_num`	µg/L	Target metal screening (Threshold: 20 µg/L)
Output Target 2	Cadmium Concentration	`Cd_num`	µg/L	Target metal screening (Threshold: 3 µg/L)

3. MODEL PREPARATION & ACCURACY PERFORMANCE

Two machine learning pipelines were trained using **Repeated 5×5 Nested Cross-Validation** to ensure zero data leakage and reliable generalization:

Target Metal	Model Type	Accuracy Score (R^2)	Prediction Error (RMSE)	Water Management Interpretation
Cadmium (Cd)	Linear Regression	**R^2 = 0.71** (71% variance explained)	0.0658 µg/L	**High Predictive Accuracy:** Reliable for quantitative estimation and field risk prioritization.
Nickel (Ni)	Huber Regressor	**R^2 = 0.22** (22% variance explained)	1.7707 µg/L	**Baseline Screening Tool:** Suitable for preliminary risk categorization, not exact concentration prediction.

4. DECISION SUPPORT & UNCERTAINTY BOUNDS (CONFORMAL PREDICTION)

To guarantee water management safety and prevent false negatives, every model prediction is wrapped with a **90% Conformal Uncertainty Interval**:

- **Observed Reliability: 90.0% coverage** (36/40 evaluation samples fell within predicted confidence bands).
- **5-State Screening Decision Categories:**
- `BELOW_THRESHOLD`: Metal concentration is reliably below regulatory limits.
- `UNCERTAIN`: Prediction interval overlaps safety threshold; laboratory re-testing recommended.
- `POTENTIAL_EXCEEDANCE`: High likelihood of exceeding safety limits; priority site action required.
- `OUT_OF_DOMAIN`: Water parameters fall outside validated regional limits (pH < 6.87 or > 7.32, TDS > 593.4 mg/L).
- `INVALID`: Inputs physically impossible (e.g., negative depth or TDS).

5. COMPLETE FILE & DATASET INVENTORY MAP

Phase	Purpose	File Name & Path	Key Contents
Raw Data	Original Laboratory Readout	`data/raw/Groundwater_quality_data_Northbengal.xlsx`	40 sample records with 26 chemical parameters
Hydrochemistry	Water Quality & Risk Indices	`data/processed/phase2_risk_indices_results.csv`	Calculated WQI, HPI, HEI, and Gibbs classifications
ML Training Set	Prepared ML Input Dataset	`data/processed/phase4_features_track1.csv`	Clean 40-row feature matrix (`pH_proxy` , `TDS_calc` , `NO3-N` , `WELL_DEPTH`)
Predictions	Final Screening Decisions	`water_final_results/predictions/FINAL_SCREENING_DECISIONS.csv`	Per-well model predictions, uncertainty intervals, & screening decisions
Audit Report	Technical Validation Report	`scientific_audit_report.pdf`	Complete 5-page forensic audit for peer review

CONCLUSION FOR WATER MANAGEMENT POLICY

This pipeline demonstrates that routine water quality parameters can serve as an **effective, low-cost first line of defense** for groundwater monitoring in North Bengal, enabling authorities to target expensive heavy-metal laboratory analysis only where screening uncertainty or risk exceedance is flagged.